MILITARY SPECIFICATION

MICROPHONE M-33A/AIC

1. SCOPE

1.1 This specification covers one type of hand-free, noise-cancelling, dynamic microphone, designated Microphone M-33A/AIC. (See 6.1.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-585—Boxes, Wood, Wirebound.
PPP-B-591—Boxes, Fiberboard, Wood Cleated.
PPP-B-601—Boxes, Wood, Cleated.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Boxes, Fiber, Solid.
PPP-T-97 —Tape; Pressure-Sensitive Adhesive Filament Reinforced.

MILITARY

MIL-P-116 —Preservation, Methods of.
MIL-B-10377 —Box, Wood, Cleated, Veneer, Paper-Overlaid.
MIL-M-13231—Marking of Electronic Items.
MIL-F-14072 —Finishes for Ground Signal Equipment.

STANDARDS

MILITARY

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.
MIL-STD-169—Extreme - Temperature Cycle.
MIL-STD-170—Moisture Resistance Test Cycle for Ground Signal Equipment.
MIL-STD-202—Test Methods for Electronic and Electrical Component Parts.

DRAWINGS

SIGNAL CORPS

SC-DL-176686 —Microphone M-33( )/AIC.
SC-A-362100 —Preparation of Preproduction Packs.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Construction. The equipment shall be constructed in accordance with Drawing SC-DL-176 686. (Column M-33A/AIC (4639-PP-61)) Exception to drawings—Replace Specification Reference MIL-M-9239 with MIL-M-55233.

3.2 Preproduction samples. The contractor shall furnish ten preproduction samples for approval, if required by the Invitation for Bid and Contract. (See 4.3 and 6.2)

3.3 Parts, materials and processes.

3.3.1 *Plastic material and parts.* Where not machined, plastic material and parts shall have the original smooth or polished

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surfaces. Surfaces that have been sawed, cut, punched, or otherwise machined shall be as smooth as practicable in accordance with good manufacturing practice for the intended application.

3.3.2 *Soldering.* Soldering shall conform to the following:

3.3.2.1 *Acid or acid salts.* No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation materials. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.3.2.2 *Process.* There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment, flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.3.3 *Cleaning.*

3.3.3.1 *Parts.* After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted.

3.3.3.2 *Units.* After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.3.4 *Finish.* The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings. (See 4.4)

3.3.5 *Marking.*

3.3.5.1 *General.* Marking shall conform to MIL-M-13231. (See 4.4)

3.3.5.2 *Visibility.* Wherever practicable, parts shall be so mounted that their identification markings shall be readily visible with minimum disassembly of the equipment.

3.4 Service conditions. The equipment shall meet the following service conditions—where a test is referenced, meeting the test shall be considered as compliance with the requirement.

3.4.1 *Operational.* Continuous use for a period of 2000 hours, under the world-wide environmental conditions specified in 3.4.2 through 3.4.8, with a duty cycle of ¼ minute on and ¾ minute off and with no more than normal maintenance and replacement of parts. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc, he shall be guided by this requirement Approval of the preproduction sample (see 3.2) shall be considered as compliance with this requirement.

3.4.2 *Temperature.* (See 4.7.1.)

(a) *Operating:* Ambient temperature in the range of +150° F to −65° F, exposure at the high temperature extreme not to exceed four hours, and the low temperature extreme not to exceed 72 hours, at any one time.

(b) *Nonoperating:* Exposure in the range of +160° F to −80° F; exposure at the high temperature extreme not to exceed four hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

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3.4.3 *Relative humidity.* (See 4.7.2) Exposure to 97 percent relative humidity for 20 hours and 100 percent relative humidity (with condensation) for four hours for each 24 hour period over the period of hours stated in 3.4.1. Meeting the tests of 4.7.2 shall be considered compliance with this requirement.

3.4.4 *Elevation.* (See 4.7.3.)

(a) *Operating:* Up to 15,000 feet above sea level. (See 4.8.4 and 4.8.6.)

(b) *Nonoperating:* Up to 50,000 feet above sea level. (See 4.7.3)

3.4.5 *Shock, drop.* (See 4.7.4.) The equipment shall show no physical damage except that minor damage is permitted. The equipment shall also meet the specified performance for the following:

Impedance	3.5.1
Sensitivity	3.5.3
Response-frequency characteristic	3.5.5
Noise discrimination	3.5.6

3.4.6 *Salt spray* (See 4.7.5.) Twenty percent salt solution at 96° F for 48 hours.

3.5 Performance requirements.

3.5.1 *Impedance.* (See 4.8.2.) The electrical impedance of the microphone at 1000 cps shall be 3 ohms ± 0.2 ohm.

3.5.1.1 *Impedance-frequency characteristic* The electrical impedance of a particular microphone of any frequency over the range of 200 cps to 7000 cps shall not deviate from the 1000-cps impedance by more than 20 percent. The impedance-frequency characteristic shall be reasonably smooth over the frequency range of 200 cps to 7000 cps.

3.5.2 *Breakdown voltage, insulation.* Electrical insulation shall withstand a minimum breakdown voltage of 500 volts root mean square (rms) at 60 cps. After a 10 second application of the voltage there shall be no visible deterioration and the microphone shall meet the specified performance requirements. (See 4.8.3.)

3.5.3 *Sensitivity.* (See 4.8.4.) The microphone output voltage in the frequency band of 750 to 1250 cps with the moisture barrier and shield in place, shall be as follows, measured across a resistive load of five ohms when a sound pressure of 74 decibels (db) relative to 0.0002 dynes per square centimeters (dyne/cm^2) is applied:

At ground level: from three to five microvolts.

At 15,000 feet: within three db of the voltage developed at ground level.

3.5.4 *Response frequency characteristic at ground level.* The response frequency characteristic of the microphone at ground level shall be such that it can be approximated by a trend line as shown in Figure I It shall be smooth and evidence no peaks or dips over the frequency range of 200 cps to 1000 cps. (See 4.8.5.)

3.5.4.1 *Maximum deviation from trend line.* The maximum deviation of the response frequency characteristics from the trend line shall not exceed the limits shown in Figure I. In the frequency range from 1200 to 4500 cps, the number of deviations which are greater than five db from the trend line shall not exceed four. (See 4.8.5.1.)

3.5.4.2 *Average deviation from the trend line.* The average deviation from the trend line shall be no greater than 1 db in the frequency range from 200 to 600 cps and no greater than 2 db in the frequency range from 600 to 4500 cps (See 4.8 5.2.)

3.5.5 *Response-frequency characteristic at 15,000 feet.* (See 4.8.6.) The response of the microphone at an altitude of 15,000 feet at any frequency over the following frequency ranges shall not differ from the response at ground level at the same frequency by an amount greater than that indicated below:

Frequency range (cps)	Deviation (db)
200–750	6
750–3500	2
3500–4500	8

3.5.6 *Noise discrimination characteristics.* The average noise discrimination character-

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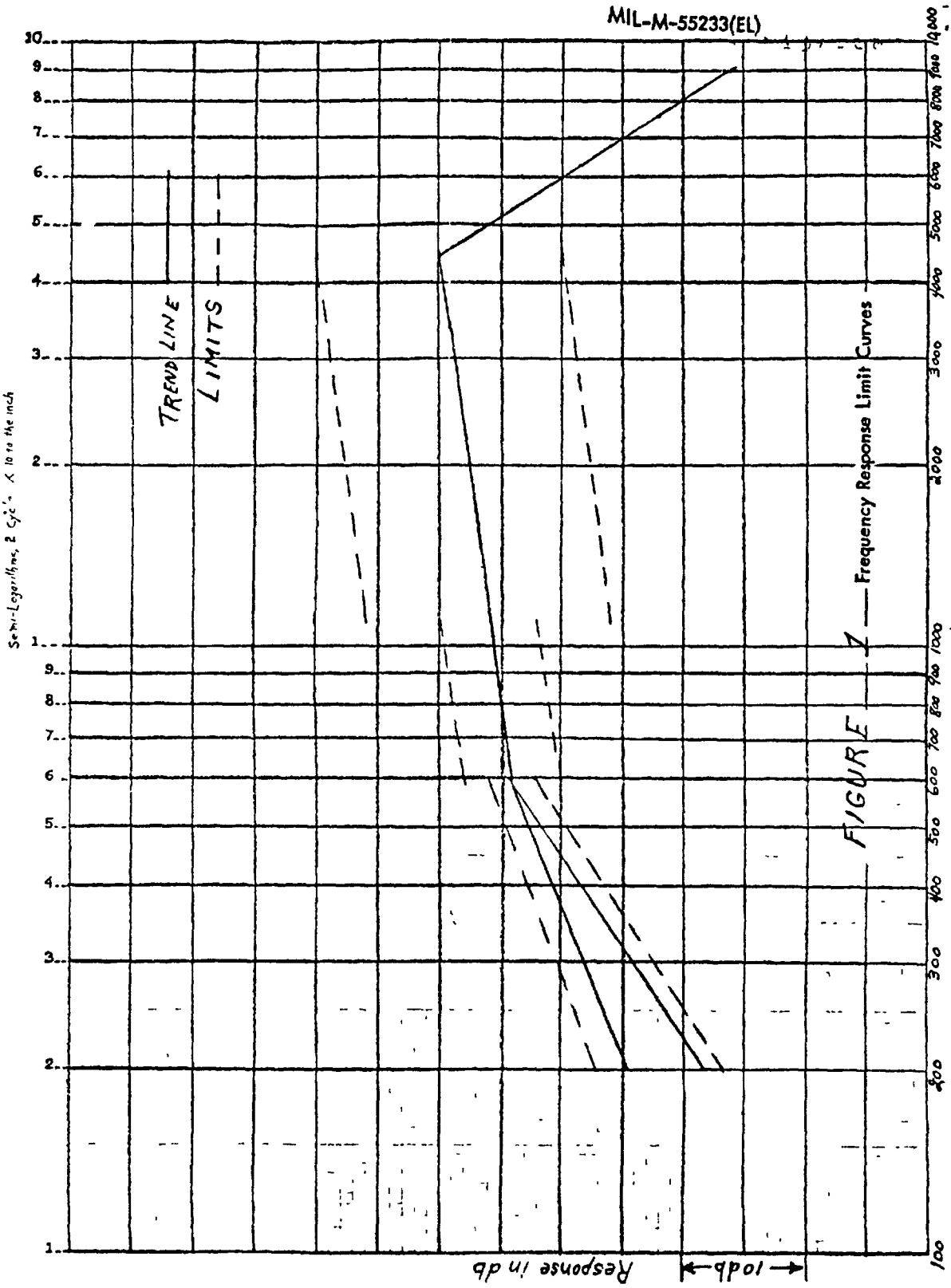


FIGURE 1 *Frequency response limit curves.*

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istic of the microphone and the peaks of the front and rear noise discrimination characteristics shall fall within an envelope four db wide, having a slope of six db per octave falling between 200 and 6000 cps. The front and rear characteristics may intersect the lower side of the envelope but shall not intersect the envelope on the top side. The average noise discrimination curve shall intersect its zero axis at some frequency between 2000 and 3000 cps. (See 4.8.7.)

3.5.7 *Total harmonic distortion.* The total harmonic distortion at any frequency within the range from 200 to 6000 cps shall not exceed six percent when measured with an input sound pressure of 134 db relative to 0.0002 dynes per square centimeter. (See 4.8.8.)

3.5.8 *Linearity.* The output of the microphone shall be a linear function of the sound pressure input to the microphone in the range of inputs from 74 to 134 db inclusive, relative to 0.0002 dynes per square centimeter (See 4.8.9.)

3.5.9 *Sensitivity to external magnetic field.* The sensitivity of the microphone to an external 400 cps magnetic field shall be no greater than one-tenth the sensitivity of a coil consisting of 123 turns of Number 34 enameled wire closely wound in a single-layer solenoid upon a nonconductive, nonmagnetic mandrel 0.500 inch in diameter, placed in the same magnetic field. (See 4.8.10)

3.5.10 *Microphone stray magnetic field.* The stray magnetic field of the microphone shall cause a magnetic compass at a distance of eight inches to deflect no more than five degrees. (See 4.8.11.)

3.5.11 *Talk-out.* There shall be no subjective difference in the output of the microphone when the microphone is tested by talking into the front and back sound parts respectively with equal loudness and with lips touching the moisture barrier shield. (See 4.8.12.)

3.5.12 *Pull and twist (cord).* The cord and its anchorage in the microphone and plug shall show no cracks, breakage, deterioration, deformity, discontinuity, shorts, loosening, or turning of the cord in its anchorage when subjected to a 15-pound pull and twist. (See 4.8.13.)

3.6 **Air gap.** (Voice coil). There shall be no foreign material present in the air gap when viewed through microscope. (See 4.9.)

3.7 **Interchangeability.** Like units, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. (See 4.10.) Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, characteristic, etc.

3.8 **Final operation.** Just prior to packaging, the equipment shall be examined visually and checked electrically as specified in 4.11

3.9 **Workmanship.** The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein:

3.3 1 General requirements for plastic material and plastic parts.
3.3.2 Process for soldering.
3.3.3 Cleaning

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Classification of inspection.** Inspection shall be classified as follows:

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(a) Preproduction inspection (does not include preparation for delivery). (See 4.3.)

(b) Inspection covered by subsidiary documents. (See 4.4.)

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5.)

(2) Quality conformance inspection of preparation for delivery. (See 4.13.)

4.3 Preproduction inspection. This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table I, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A, group B, (see tables II, III, respectively). The preproduction inspection will normally be performed in this order: shock, drop, other preproduction inspection may precede, follow, or be interspersed between the foregoing.

TABLE I *Preproduction inspection*

Inspection (For additional preproduction inspection see 4.3)	Req para	Insp para
Temperature	3.4.2	4.7.1
Moisture resistance	3.4.3	4.7.2
Elevation	3.4.4	4.7.3
Shock, drop	3.4.5.2	4.7.4
Salt spray	3.4.7	4.7.5

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	*Where required*
Finish	3.3.4
Marking	3.3.5

4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government, at its discretion, may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.6) Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 *Group A inspection.* This inspection, including sampling, shall conform to table II and the ordinary inspection procedures of Standard MIL-STD-105 Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection (4.11) shall be last. (See 6.5.)

4.5.2 *Group B inspection* This inspection, including sampling, shall conform to table III and to the special procedures for small-sample inspection of Standard MIL-STD-105. The reduced inspection procedure shall be R-1. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

4.6 Quality conformance of preparation for delivery. Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of section 5.

4.7 Service conditions tests.

4.7.1 *Temperature test.* (See 3.5.2.) The equipment shall be subjected to the temperature cycle shown on Standard MIL-STD-169. Performance shall be as follows:

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Steps	*Paragraph*	*Measurement*	*Performance*
1, 3, 7 and 10	4.8.2	Impedance	As specified in 3.5.1 with 0.25 ohm maximum degradation
	4.8.4	Sensitivity	As specified in 3.5.3 with 2 db maximum degradation.
	4.8.5	Response-frequency	As specified in 3.5.4 with 3 db maximum degradation over frequency range of 500 to 3500 cps and 5 db maximum over frequency ranges of 200 to 500 cps and of 3500 to 4500 cps.

TABLE II. *Group A inspection*

Inspection	Req Para	Insp Para	AQL	
			Major	Minor
Visual and mechanical	3.9	4.12	1%	4%
Air gap	3.6	4.9	1%	(*)
Electrical				
Impedance	3.5.1	4.8.2		
Breakdown voltage, insulation	3.5.2	4.8.3		
Sensitivity (close talking)	3.5.3	4.8.4		
Response-frequency (ground level)	3.5.4	4.8.5		
Maximum deviation	3.5.4.1	4.8.5.1	1 dphu **	(*)
Average deviation	3.5.4.2	4.8.5.2	for	
Response frequency at 15,000 feet	3.5.5	4.8.6	the	
Noise discrimination	3.5.6	4.8.7	group	
Total harmonic distortion	3.5.7	4.8.8	combined	
Linearity	3.5.8	4.8.9		
Sensitivity to external magnetic field	3.5.9	4.8.10		
Microphone Stray magnetic field	3.5.10	4.8.11		
Talk-out	3.5.11	4.8.12		
Pull and twist (cord)	3.5.12	4.8.13		
Final operation	3.8	4.11	1%	

* All electrical, air gap and final operation defects are considered major
** Defects per hundred units

TABLE III. *Group B inspection*

Inspection	Req para	Insp para	AQL	Inspection level	
				Normal and tightened	Reduced
Dimensional interchangeability	3.7	4.10	6.5%	L-7	L-5

4.7.2 *Moisture resistance test.* (See 3.4.3.) The equipment shall be tested as follows:

4.7.2.1 *Test method.*

(a) Dry at 130° + 5° F for 24 hours.
(b) Condition at 77° ± 5° F and 40 to 50 percent relative humidity for 24 hours.
(c) Take measurements as specified in 4.7.2.2.
(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170.
(e) After cycling has been completed, condition the equipment for 24 hours at 77° ± 5° F and 40 to 60 percent relative humidity. The equipment shall meet the performance specified in 4.7.2.2.

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4.7.2.2 *Performance.* Measurements made after test and resulting performance shall be as follows:

Paragraph	*Measurement*	*Performance after cycling*
4.8.2	Impedance	As specified in 3.5.1 with 0.25 ohm maximum degradation.
4.8.4	Sensitivity	As specified in 3.5.3 with 2 db maximum degradation
4.8.5	Response-frequency	As specified in 3.5.4 with 3 db maximum degradation over frequency range of 500 to 3500 cps and 5 db maximum over frequency ranges of 200 to 500 cps and 3500 to 4500 cps.

4.7.3 *Elevation test.* (See 3.4.4.) The equipment shall be tested as follows:

(a) With the equipment stabilized at room temperature, take the following measurements:

Paragraph	*Measurement*
4.8.4	Sensitivity
4.8.5	Response-frequency

(b) With the equipment placed in its normal operation position in an altitude chamber, reduce the barometric pressure to 3.4±0.1 inches of mercury (simulated 50,000 feet above sea level). Maintain this condition for one hour.

(c) Return the chamber to ambient pressure. After stabilization, repeat the measurements specified in 4.7.3(a). The equipment shall meet specified performance for those measurements, except for the following degradation, sensitivity, 2 db maximum; response-frequency, 3 db maximum over frequency range of 500 to 3500 cps and 5 db maximum over frequency ranges of 200 to 500 cps and of 3500 to 4500 cps.

4.7.4 *Shock test: drop.* (See 3.4.5.) The microphone shall be subjected to a total of 26 random drops from a height of 48 inches. Drops shall be made from a quick-release hook; or pendulum tester as made by the L. A. B. Corporation, Skaneateles, New York, or equal. The floor or barrier receiving the impact shall be of two-inch fir backed by concrete or a rigid steel frame.

4.7.5 *Salt-spray test.* (See 3.4.7.) The microphone shall be tested in accordance with Condition B, Method 101 of Standard MIL-STD-202. A 20 percent salt solution for 48 hours shall be used. After completion of the test, the microphone shall be washed and wiped or shaken dry. The sensitivity and response frequency (see 4.8.4 and 4.8.5, respectively) shall be measured at the completion of the test and again 24 hours later

4.8 **Performance tests.**

4.8.1 *Test equipment.* The following test equipment shall be used for conducting the tests specified herein:

(a) Condenser microphone: The microphone used as a primary standard shall be Western Electric Model 640-AA, or equal, which has certified free field and pressure calibrations.

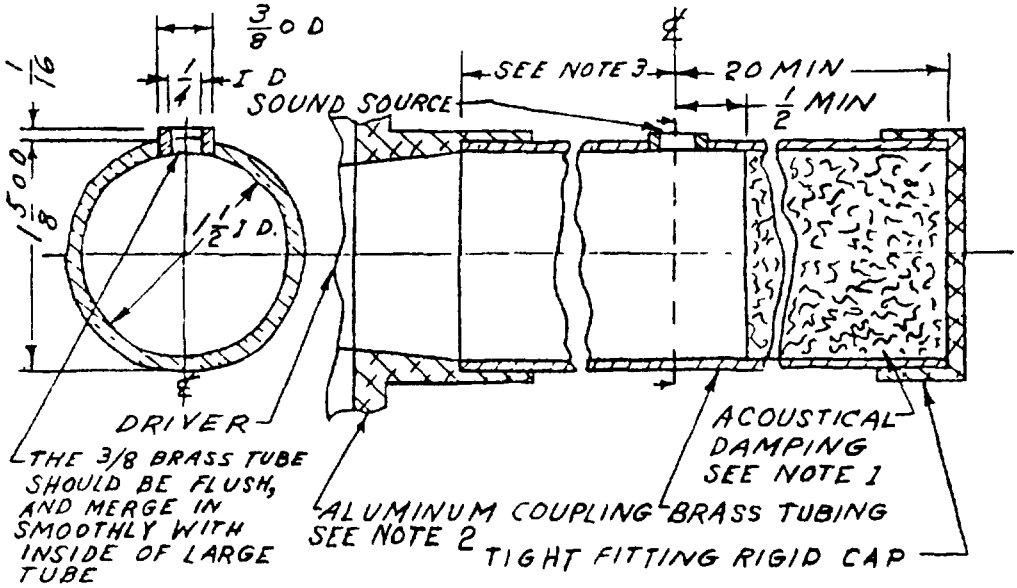
(b) Audio oscillator: The audio oscillator shall have a frequency range of at least 20 to 20,000 cps, a total harmonic distortion at any frequency of not more than one percent, and an accuracy with ± one percent ±0.5 cycle.

(c) Vacuum tube voltmeter (VTVM): The vacuum tube voltmeter shall have a flat frequency response (±1 db) from 20 to 20,000 cps and shall be capable of measuring voltages from 0.001 volt to 10 volts, within an accuracy of two percent. The input impedance shall be at least 500,000 ohms.

(d) *Calibrated close-sound source:* The close-sound source used shall con-

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form to figure 2 and shall be capable of producing a sound pressure level of 100 dynes per square centimeter at an axial distance of ¼ inch from the mouth opening with no more than five percent total harmonic distortion over a frequency range from 200 to 10,000 cps. It shall be calibrated at each altitude at which it is used, with a pressure calibrated standard microphone as specified in 4.8.1(a) placed ¼-inch from, and coaxial with, the front face of the sound source opening with the plane of the diaphragm of the standard microphone normal to the axis of the source. There shall be no added baffles or reflecting surfaces in the vicinity of the test position or voice opening during calibration.



1. The end of the tube shall be filled evenly with an acoustical damping material, such as ozite, so as to provide a resistive termination equal to the characteristic impedance of the tube. The damping material shall begin at a point not less than one half inch beyond the sound source orifice. The length of the damped portion of the tube shall be not less than 20 inches. Approximately 45 grams of ozite are required in a 20 inch length of tube.

2. If the throat of the driver is unequal in diameter to the voice tube, a smooth transition shall be made, so as to match properly the characteristic impedance of the tube to driver mechanism.

3. This dimension is non-critical and may be varied up to a length of one foot to suit the needs of the microphone mounting fixtures.

4. All dimensions in inches. Unless otherwise specified, tolerances! ± 1/64
Scale: Full

FIGURE 2 *Artificial voice (¼ dia "close" sound source)*

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(e) Distortion analyzer: The distortion analyzer shall have a frequency range of at least 40 to 10,000 cps, and shall measure residual harmonic frequencies within ± 3 percent for distortion levels as low as ½-percent.

(f) Magnetic compass: The magnetic compass shall be a Keuffel and Esser Number 5600 Forestry Compass, or equal.

(g) Field coil: The field coil shall consist of a coil of 700 turns of Number 17 enameled wire six inches by six inches in cross section or six inches in diameter with a winding length of ten inches.

(h) Calibrating coil: The calibrating coil shall consist of a single-layer solenoid wound with 123 turns of Number 34 enameled wire on a non-conductive, non-magnetic mandrel 0.5 inches in diameter.

(i) Area measuring instrument (planimeter or equivalent): The instrument shall have an accuracy within three percent of actual area measured.

(j) Audio amplifier: The audio amplifier shall have a frequency response flat to within ± two db from 100 to 15,000 cps. It shall incorporate no automatic gain control or other circuits which might tend to mask deficiencies of the microphone.

(k) Headset: The headset shall have a frequency response flat to within ± three db from 100 to 5000 cps

4.8.2 *Impedance.* Using the audio oscillator specified in 4.8.1(b) as a signal source, a 1000 signal shall be delivered through a series non-inductive resistor of 10 ohms ± 2 percent to the microphone terminals. The voltage across the microphone terminals as measured with the VTVM specified in 4.8.1(c) shall not exceed 0.005 volts. The voltage across the series resistor shall be measured using the same VTVM. The impedance of the microphone shall be calculated using the following formula:

$$Z = \frac{E_1 R}{E_2}$$

where: Z = Impedance of microphone
E_2 = Voltage across series resistance
E_1 = Voltage across microphone
R = Series resistor

The above test shall be repeated using a signal source of 200, 500, 2000, 5000, and 7000 cps respectively. The results shall conform to 3.5.1.

4.8.3 *Breakdown voltage test.* With each microphone terminal stud connected to the voice coil lead, a 500 volt rms, 60 cps alternating potential shall be applied between each terminal stud and each of the metal parts to the studs, voice coil, and voice coil leads. The 500 volt potential shall be maintained for at least ten seconds. The microphone shall be observed for evidence of breakdown and flash over. The results shall conform to 3.5.2.

4.8.4 *Sensitivity test.* The calibrated close sound source specified in 4.8.1(d) shall be used to impress a series of three input signals upon the microphone at a sound pressure level of 74 db relative to 0.0002 dynes/cm², as follows:

750 cps
1000 cps
1250 cps

At each frequency, the output voltage developed by the microphone across a five-ohm resistive load shall be measured on the VTVM (see 4.8.1(c)) and recorded. The three voltage measurements shall be averaged and the resultant voltage recorded.

The above test shall be repeated with the microphone and calibrated sound source enclosed in a chamber in which the barometric pressure is reduced to 16.9 inches of mercury (simulated 15,000 feet above sea level).

The results shall conform to 3.5.3. (Also see 3.4.4.)

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4.8.5 *Response frequency characteristics tests.* The close sound source (see 4.8.1(d)) shall be used to impress signals provided by the audio oscillator (see 4.8.1(b)) of 200, 300, 400, 600, 1000, 1200, 1500, 1800, 2000, 3000, 3500, 4000, 4500, 5000, 6000 cps respectively upon the microphone at a constant input sound pressure. The output voltages developed by the microphone across a five-ohm resistive load shall be measured on the VTVM (see 4.8.1(c)) and recorded. The response-frequency characteristic shall be plotted as a connected curve on semi-logarithmic paper with frequency displayed on the logarithmic abscissa scale. The linear ordinate scale shall be recorded in db with a scale of 10 db per inch. The characteristic shall be compared with the trend line shown in figure 1. The results shall conform to 3.5.4.

4.8.5.1 *Maximum deviation from trend line.* The trend line and limit line shown in figure 1 shall be superimposed on the response-frequency characteristic determined in 4.8.5 in the most favorable position to determine conformance to 3.5.4.1.

4.8.5.2 *Average deviation from trend line.* The arithmetic sum of the areas bounded by the response-frequency characteristic and the trend line between 200 cps and 600 cps (see 4.8.5.1) shall be determined using the area measuring instrument specified in 4.8.1(i). Repeat the above measurement for the portion of the curves between 600 cps and 4500 cps. Resultant areas no greater than the areas of the following rectangles shall be considered as conformance to 3.5.4.2.

Measured portion	*Length*	*Height*
200 cps to 600 cps	Length of trend line	1 db on response scale
600 to 4500 cps	Length of trend line	2 db on response scale

4.8.6 *Response-frequency characteristic at 15,000 feet.* Repeat the test specified in 4.8.5 with the microphone and close sound source enclosed in a chamber in which the barometric pressure is reduced to 16.9 inches of mercury (simulated 15,000 feet above sea level). The results shall conform to 3.5.5. (Also see 3.4.4.)

4.8.7 *Noise discrimination.* The microphone shall be tested for comformance to 3.5.6 as follows:

(a) Measure the sensitivity to a close sound source at a frequency of 1000 cps as specified in 4.8.4.

(b) Measure the response-frequency characteristic to a close sound source as specified in 4.8.5.

(c) Measure the sensitivity as specified in 4.8.4 except that it shall be to a distant sound source (microphone placed in a calibrated sound field at least six feet from the sound source in an anechoic chamber) at 1000 cps.

(d) Measure the response-frequency characteristic as specified in 4.8.5 except that it shall be to a distant sound source (see (c) above).

(e) The distant sensitivity and distant response-frequency shall be measured twice; once with the front sound port (front distance response) facing the sound source and once with the rear sound port (rear distance response) facing the sound source. In each case, the microphone shall be oriented for maximum output at 1000 cps.

(f) Plot the close response-frequency on semi-logarithmic coordinate paper as specified in 4.8.5.

(g) Plot the front distance response curve and the rear distance response curve on the same coordinates, with the point on the curve at 1000 cps located below the point on the close response curve

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at 1000 cps by the number of db equivalent to the ratio of close sensitivity at 1000 cps in volts to the distant sensitivity at 1000 cps in volts; ie,

$$\text{number of db} = 20 \log_{10} \frac{\text{close sensitivity at 1000 cps}}{\text{distant sensitivity at 1000 cps}}$$

(h) The two curves, front distance response and rear distance response, shall be individually substracted from the close-response curve and the two difference curves plotted (front and rear noise discrimination characteristics, respectively).

(i) An average of the two difference curves shall be plotted on the same coordinates and examined for conformance to 3.5.6.

4.8.8 *Total harmonic distortion.* With the microphone sealed (air-tight acoustic seal) to the close sound source, measure the distortion for 134 db of sound pressure at 200, 400, 600, 1000, 2000, 3000, 4500, and 6000 cps using the distortion analyzer specified in 4.8.1(e). Results shall conform to 3.5.7.

4.8.9 *Linearity.* Using sound pressure inputs of 74 db, 94 db, 114 db, 124 db and 134 db relative to 0.0002 dynes/cm^2, measure the output voltage developed by the microphone across a five-ohm resistive load for each input at 500, 1500, and 3000 cps. Plot the results on a rectangular graph paper and examine for conformance to 3.5.8.

4.8.10 *Sensitivity of external magnetic field.* The field coil specified in 4.8.1(g) shall be excited with 250 milliamperes of 400 cps alternating current. The VTVM specified in 4.8.1(c) shall be connected to the microphone with the leads from the microphone to the voltmeter closely twisted. The microphone shall then be oriented in the center of the field coil for maximum output voltage. The voltage indication shall conform to 3.5.9.

4.8.11 *Microphone stray magnetic field.* Using a magnetic compass specified in 4.8.1(f), place the microphone so that the geometric center of the motor is eight inches from the pivot point of the compass needle and on the perpendicular bisector of the needle in the plane of rotation of the needle. The microphone shall be oriented in all directions and the maximum deflection of the compass observed. The results shall conform to 3.5.10.

4.8.12 *Talk-out test* The microphone shall be connected through an amplifier specified in 4.8.1(j) to a headset specified in 4.8.1(k). The tester shall talk separately into the front and back sound ports of the microphone with equal loudness and with lips touching the moisture barrier shield. The tester shall listen to his own voice to determine if there is a difference in the general quality of reproduction when he is talking in either the front or back sound port. The results shall conform to 3.5.11.

4.8.13 *Pull and twist (cord).* The cord and its anchorage in the microphone and plug shall be subjected to a 15 pound pull for a period of 30 minutes. During the last five minutes of the test, and with the 15 pounds applied, twist the cord three full turns in each direction from the neutral position within a length of six inches from the anchorage at each end of the cord so that the torque is applied to the anchorage The results shall conform to 3.5.12.

4.9 Air gap. The air gap shall be examined through a microscope having a magnification of at least 15 diameters for conformance to 3.6.

4.10 Inspection for dimensional interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.7. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

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(a) Internal dimensions of the case and external dimension of insertable assemblies where such dimensions affect mating of parts.

(b) Dimensions of the moisture barrier, moisture-barrier shield and wire carriage of the microphone.

4.11 Final operational test. Prior to packaging, the equipment shall be checked visually for damaged parts, loose parts and completeness A simple talk-test through on amplifier and speaker shall be performed to assure that equipment is operating. (See 3.8.)

4.12 Visual and mechanical inspection. The microphone shall be examined for defects listed in table IV.

TABLE IV *Classification of visual and mechanical defects*

Classification	Defect
Major	Centering of voice coil in air gap
	Security of microphone terminal connections to cable terminals.
Minor	Operation of microphone hinge
	Operation of fastening devices.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* The M-33A/AIC Microphone shall be preserved, packaged and tested in accordance with the applicable provisions of MIL-P-116 and in a manner that will afford adequate protection against corrosion, deterioration and damage during worldwide shipment, handling and open storage.

5.1.2 *Level C.* The M-33A/AIC Microphone shall be afforded preservation and packaging in accordance with the suppliers' normal commercial practice

5.2 **Packing.** The M-33A/AIC Microphone packaged as specified, shall be packed in shipping containers conforming to the requirements of the specifications referenced below for the designated level The gross weight shall not exceed the weight shown unless on individually packed item exceeds that amount Closure and strapping shall be as prescribed in the applicable box specification or appendix thereto, except that bands of reinforced, pressure-sensitive adhesive tape, ½ inch wide and conforming to Specification PPP-T-97 for Grade IV, shall be used for fiberboard boxes in lieu of metal strapping

5.2.1 *Level A.*

BOX SPECIFICATION		*MAX GROSS WT*
Fiberboard (Type I, Class 2)	PPP-B-636	Table VI
Nailed Wood (Style 4, Class 2)	PPP-B-621	200 lbs.
Wirebound (Style 2 or 3, Class 2)	PPP-B-585	200 lbs.
Wood Cleated Fiberboard (Overseas)	PPP-B-591	200 lbs.
Wood Cleated Plywood (Overseas) ...	PPP-B-601	200 lbs
Wood Cleated, Veneer, Paper Overlaid (Overseas)	MIL-B-10377	200 lbs

5.2.2 *Level B.*

BOX SPECIFICATION		*MAX GROSS WT*
Fiberboard (Type I, Class 1) .	PPP-B-636	Table I
Nailed Wood (Style 4, Class 1)	PPP-B-621	200 lbs
Wirebound (Style 3, Class 1)	PPP-B-585	200 lbs.
Wood Cleated Fiberboard (Domestic)	PPP-B-591	200 lbs.
Wood Cleated Plywood (Domestic) .	PPP-B-601	200 lbs.
Wood Cleated, Veneer, Paper Overlaid (Overseas)	MIL-B-10377	200 lbs

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5.2.3 *Level* C. The M-33A/AIC Microphone shall be packed in shipping containers of the type, size and kind commonly used for the purpose in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the rules or regulations of the common carrier, applicable to the mode of transportation.

5.2.4 *Pilot pack.* When Level "A" packaging and packing is specified above, one acceptable model of the M-33A/AIC Microphone will be pilot packed in a manner that will pass the preproduction tests prescribed in MIL-P-116. Illustrations of the packaging and packing procedure, together with Bills of Material, will be prepared and furnished in accordance with the provisions of Signal Corps Drawing No. SC-A-362100.

5.3 **Marking.** In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of MIL-STD-129.

6. NOTES

6.1 **Intended use.** Microphone M-33A/AIC is intended for use as a part of Headset-Microphone H-101( )/U. (See 1.1.)

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.
(b) Level of packaging and level of packing required for shipment.
(c) Preproduction inspection:
    Preproduction pack(s) as follows:
    Makeup of pack(s).
    Number of each kind of pack to be submitted.
    Inspection to be performed thereon.
(d) Marking and shipping of samples.
(e) Place of final inspection.

6.3 **Location of operational inspection.** It is desirable that the operational inspection (4.11) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.4 **Verification inspection.** Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make minimum utilization of the contractor's quality control system and the quality history of the product.

Notice: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Project No. 5965-A024

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119 R004

INSTRUCTIONS

This sheet is to be filled out by personnel of Government or contractor activities involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay, and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF YES, IN WHAT WAY?

4 REMARKS *Attach any pertinent data which may be of use in improving this specification. If there are additional papers attach to form and place both in an envelope addressed to preparing activity.*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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